



UX RESEARCH & DESIGN SYSTEM

CAIRO METRO NAVIGATION APP



METROM

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Phase 1 — UX Research

1. Survey

GOAL

Understand how users navigate Cairo Metro, what problems they face, and what they need from the app.

SURVEY QUESTIONS

- How often do you use Cairo Metro?
- What is the biggest problem you face when using the metro?
- Do you usually know your route before traveling?
- Do you prefer searching by station name or by map?
- Do you need fare information before the trip?
- Do you use mobile apps while traveling?
- What would make a metro app easier for you to use?
- Would you like live route guidance during your trip?

FINDINGS

Users want fast route planning
Many users need simple station selection
Fare information is important before the trip
Mobile-friendly design is necessary
Users prefer clear directions with minimal steps

2. Empathy Map

USER TYPE: CAIRO METRO PASSENGER

Think "I need to find the fastest route." "I don't want to waste time searching."	Feel Frustrated when routes are confusing. Anxious when changing lines. Relieved when the app is simple.
Say "Which station should I take?" "How much will the trip cost?" "Is this the right line?"	Do Search for station names. Compare routes. Check fare details. Use the app while traveling.
Pain Points Too many steps to find a route. Confusing station names. Lack of live guidance. Hard to understand transfers.	Needs Simple route planner. Clear line colors. Easy station selection. Fare and direction info.

3. Persona

Ahmed University Student, Age 22 — Cairo

Age	22
Occupation	University student + part-time worker
Location	Cairo, Egypt
Frequency	Daily metro user
Device	Mobile-first

GOAL

Travel quickly from one station to another without confusion — find the correct line, station, and fare in seconds.

BACKGROUND

Ahmed uses Cairo Metro daily for university and part-time work. He needs a fast, reliable way to check the correct line, station number, and fare without switching between multiple apps or asking strangers.

FRUSTRATIONS

- He does not always know which line to take.
- He wastes time switching between apps or asking others for directions.
- He wants an app that is optimised for mobile and works well in crowded conditions.

NEEDS

- Fast and clear route search
- Metro line colors for quick visual recognition
- Simple step-by-step instructions
- Live map or route preview during travel

Phase 2 — Design System

1. Typography

Headings Poppins / Montserrat Bold, modern, high impact. Gives a professional transportation app feel.	Body Text Inter / Roboto Clean and readable on mobile screens. Minimal and clear.
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2. Color Palette

Swatch	Color	Usage
#1e6b42	Dark Green	Main background, headers, primary brand color
#2d9e60	Mid Green	Active buttons, interactive elements
#4caf7d	Light Green	Highlights, success states
#f5c842	Yellow/Gold	Fares, warnings, important actions
#ffffff	White	Main text on dark backgrounds
#8a8a85	Gray	Secondary text, disabled states

Dark green matches Cairo’s visual identity. Yellow draws attention for fares and warnings without overuse. High contrast between text and background improves readability in transit conditions.

3. Buttons

Button Label	Type	Purpose
Find Route	Primary	Main CTA — route search trigger
Start Traveling	Primary	Initiates live trip guidance
View Fare	Accent	Highlights fare / cost info
Reset	Secondary	Clears form, lower visual weight

- Rounded corners (border-radius 24px) for a modern, approachable feel.
- Large tap targets — minimum 44px height for comfortable mobile use.
- Clear action hierarchy: green for primary, yellow for financial, gray/outlined for secondary.

4. Icons

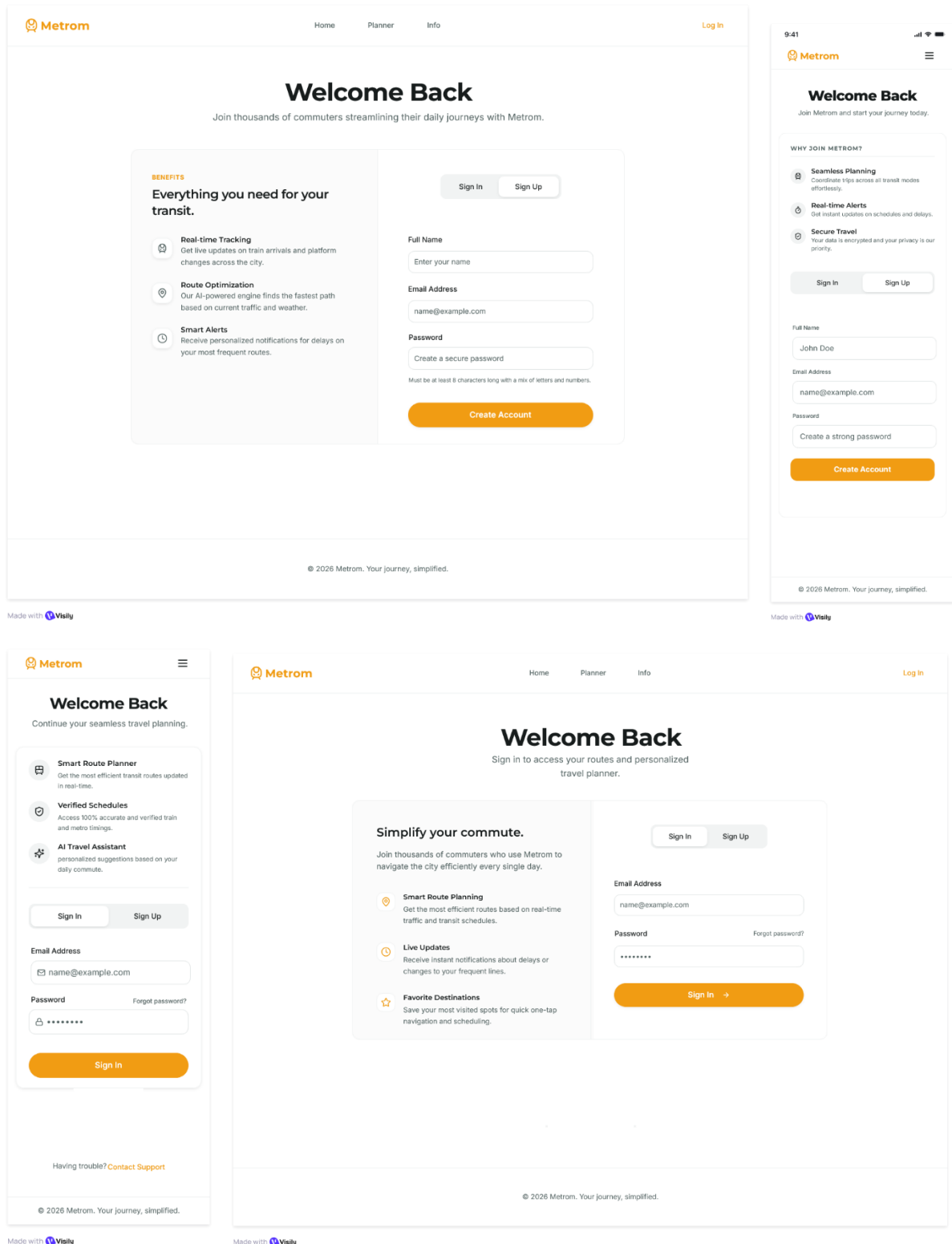
Icon	Purpose
Station icon	Identifies metro stations on the map and in dropdowns.
Search icon	Triggers the station or route search field.
Route icon	Represents the planned path between two stations.
Fare /Ticket icon	Indicates fare information and ticket pricing.
Location pin	Shows the user's current position or a selected station.
Train icon	Represents metro lines and live travel mode.

5. Wireframes

Three key screens define the core user journey in Metrom. Each wireframe below describes the layout elements and their purpose.

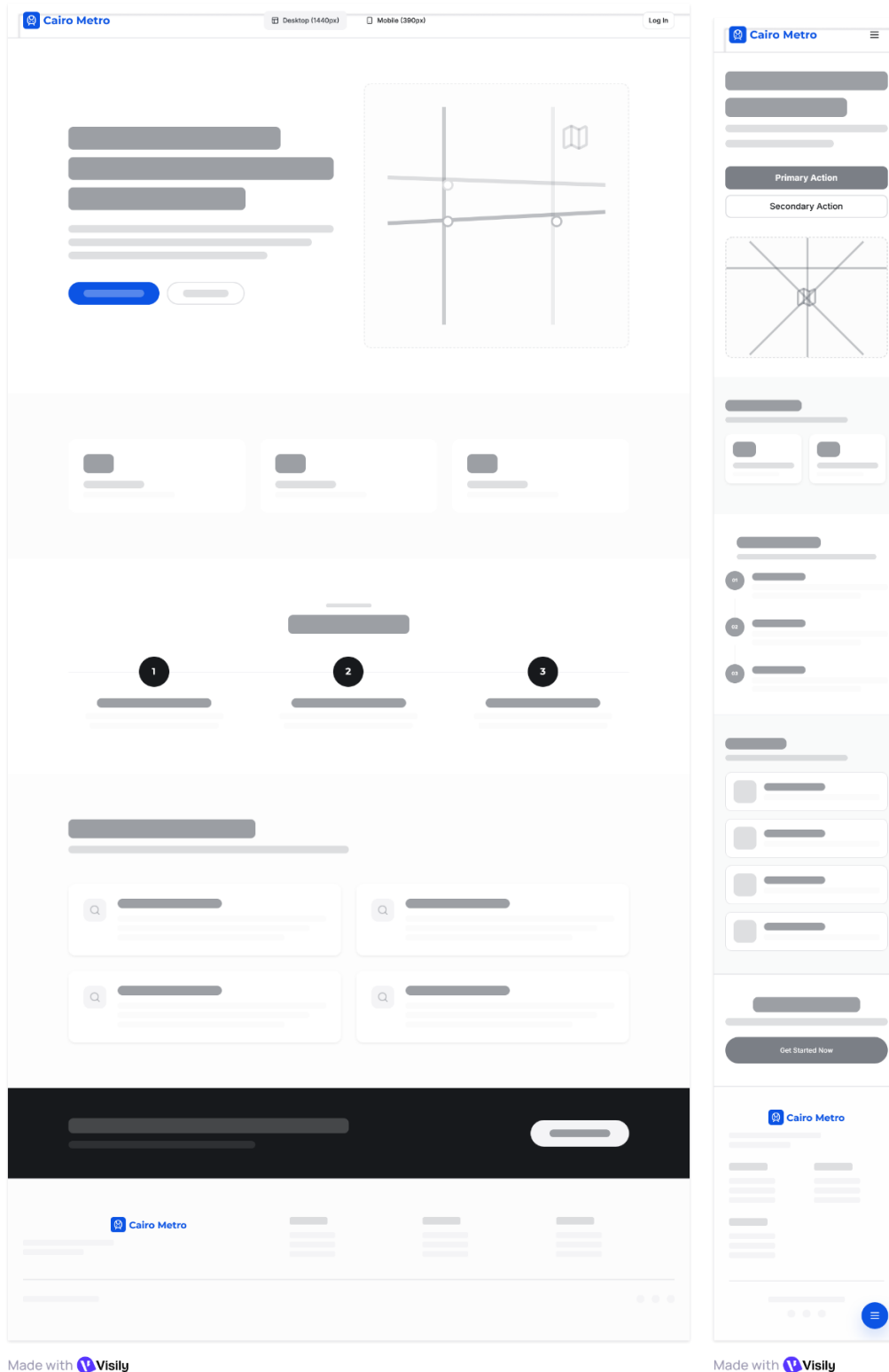
1. Sign-up / Log-in Page

This page is the entry point for new and returning users. It provides separate sign-in and sign-up states, with fields for authentication and a short section that highlights the app's benefits. The layout is designed to build trust, guide the user quickly into the app, and keep the action area simple and focused.



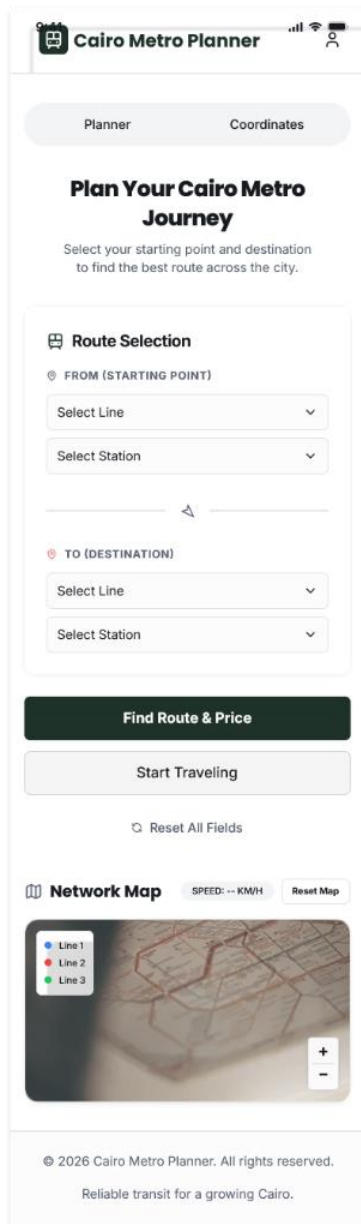
2. Home Page

The home page is the landing screen after authentication or when a visitor opens the app. It introduces Metrom, presents a short overview of the service, and links users to the main features such as planning, metro information, and station guidance. It acts as the central navigation hub and gives users a quick understanding of what the app offers.

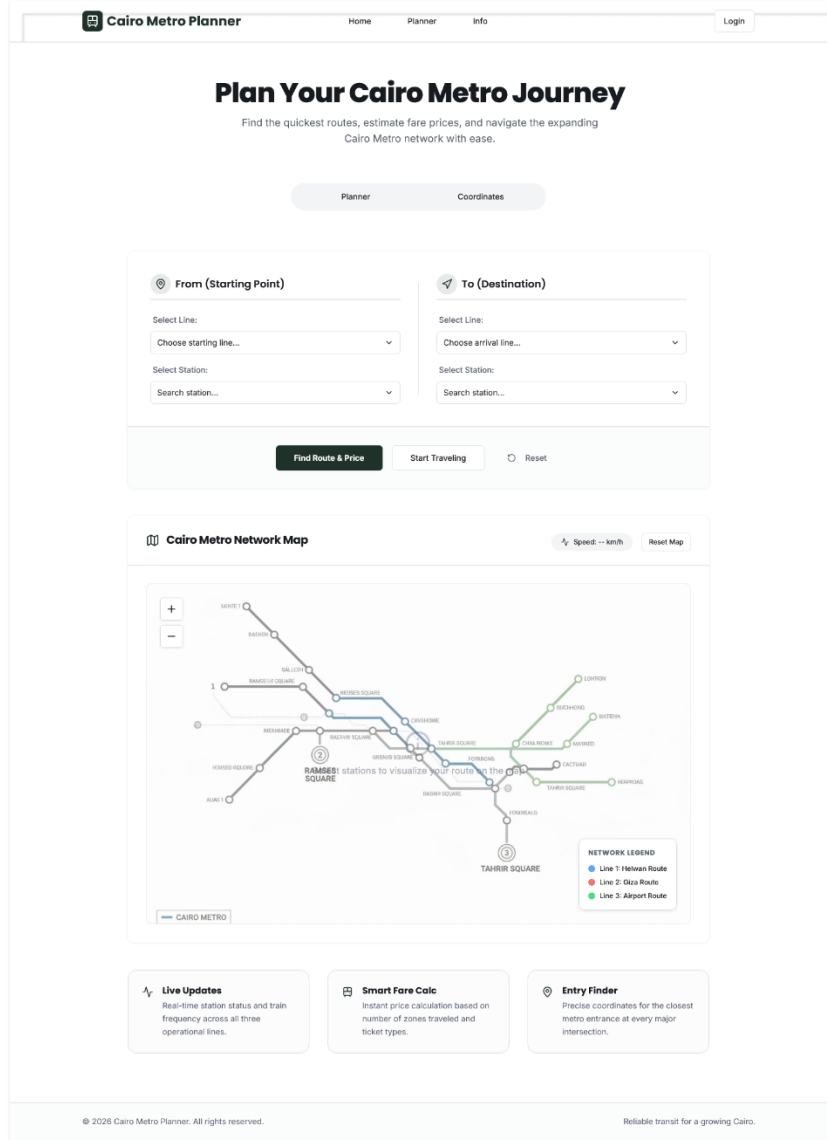


3. Planner Page

The planner page is the main trip-planning workspace. It allows users to choose their starting line and station, destination line and station, and then calculate the best route and fare. This page also includes route visualization, travel actions, and supporting metro map content, making it the core functional screen of the app.



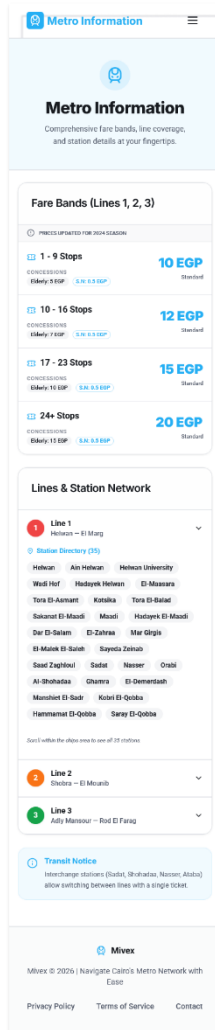
Made with Visily



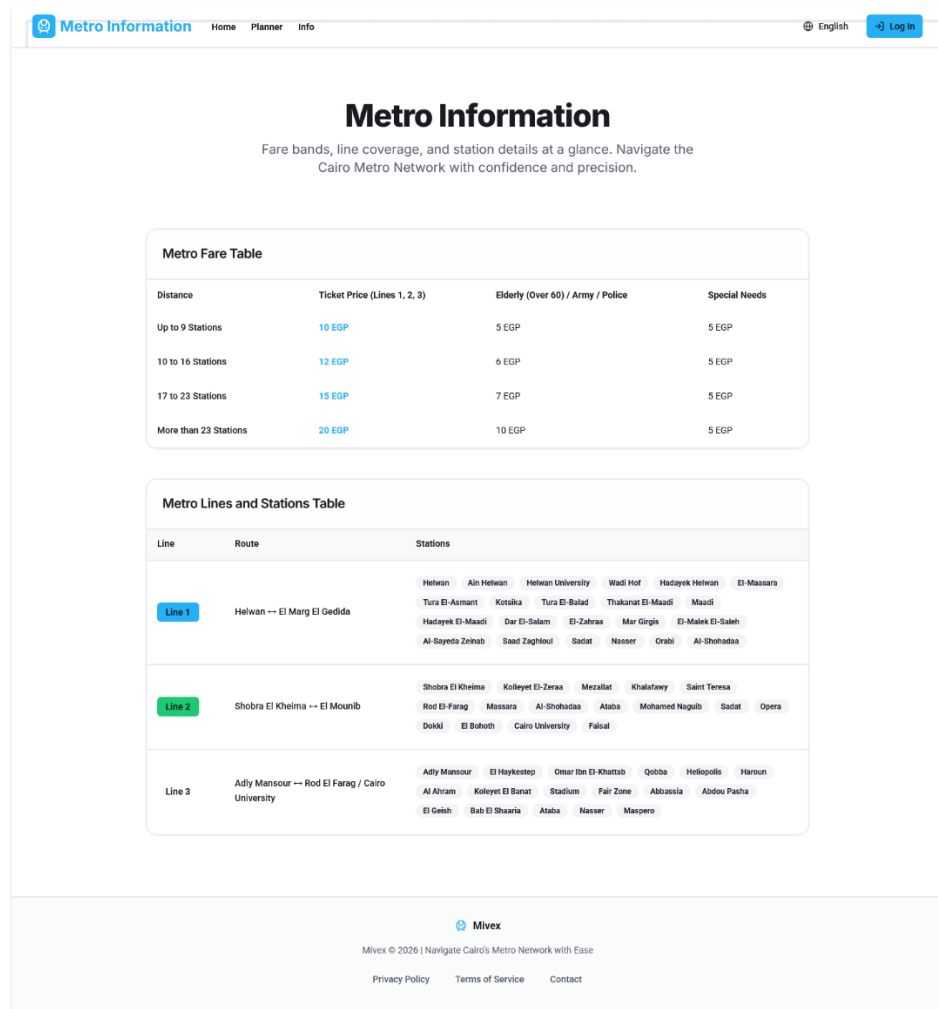
Made with Visily

4. Information Page

The information page displays static metro reference data in a structured format. It includes fare bands, line coverage, and station lists so users can quickly check prices and network details without planning a trip. This screen supports users who only need metro information and not route calculation.



Made with Visily



Made with Visily

5. Coordinates page

The coordinates page helps users find nearby metro stations using location data or map links. It is meant for cases where the user does not know the station name but knows their current location or destination coordinates. The page combines coordinate input, map-related tools, and nearest-station results to support location-based trip planning.

Cairo Metro Planner

HomePlannerInfo

Login

Plan Your Cairo Metro Journey

Pinpoint the nearest stations using geographic coordinates or map links for ultimate precision.

PlannerCoordinates

Find Closest Station by Coordinates

Enter your current coordinates and destination coordinates, or use live location for instant results.

CURRENT LOCATION

LOCATION LINK

Paste Google Maps / Waze / Apple Maps link

LATITUDE

LONGITUDE

e.g. 30.0444

e.g. 31.2357

DESTINATION LOCATION

LOCATION LINK

Paste destination map link or lat,long

LATITUDE

LONGITUDE

e.g. 30.0444

e.g. 31.2357

Import Coordinates From Links

Use Live Location

Find Closest Start & End

Closest Station To Me

Closest-station results will appear here after search.
We'll show you the nearest line, station name, and distance.

Cairo Metro Network Map

Visual network overview and real-time positioning.

STATUS: Network Active

SPEED: -- km/h

Reset Map

© OpenStreetMap contributors © CARTO

Live Network Connectivity Verified

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Reliable transit for a growing Cairo.

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Cairo Metro Planner

Metro Coordinates

Find the closest Cairo Metro stations using precise geographic location data.

PlannerCoordinates

Closest Station Lookup

Enter map links or manual coordinates below.

ORIGIN (CURRENT)

Location Link

Paste Google/Waze link...

Latitude

Longitude

e.g. 30.0444

e.g. 31.2357

DESTINATION

Location Link

Paste destination link...

Latitude

Longitude

e.g. 30.0100

e.g. 31.2000

Import from Links

Use Live Location

Closest Station To Me

Find Closest Start & End

CALCULATION SUCCESSFUL

Sadat Station

Line 1 & 2 Interchange

0.8 km

Dokki Station

Line 2 (Red Line)

1.4 km

CALCULATED IN 42MS BASED ON GPS

Transit Context Map

SPEED: -- KM/H

LIVE GPS

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6. Profile

The profile page is the user account management area. It lets users update personal details, manage saved trips, and access account-related actions such as sign out or delete account. It is designed as a personal control center where users can review their data and maintain their journey history.

PlanWise

User Name  Sign Out

Your Profile

Manage your account settings and view your historical travel itineraries.

Account Details

Update your personal information and security credentials.

Alex Harrison

alex.h@planwise.io

Min. 8 characters

Delete Account

Save Changes

Saved Trips

Your recent travel plans and generated itineraries.

EXPORT:

JSON

CSV

GeoJSON

Coastal Highway Adventure

San Francisco → Los Angeles • 452 miles

MAY 12, 2024

Alpine Lakes Loop

Geneva → Interlaken → Zurich • 185 miles

APR 28, 2024

Tokyo City Explorer

Shinjuku → Shibuya → Asakusa • 12 miles

MAR 15, 2024

View all 12 historical trips

Your data is encrypted and secure. PlanWise does not sell your location history to third parties. Need help?Contact Support

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9:41

PlanWise

User Name 

Your Profile

Manage your account settings and trip data.

Account Details

Update your personal information and credentials.

Alex Johnson

alex.j@example.com

Leave blank to keep current

Save Changes

Delete Account

Saved Trips

4 Total

EXPORT DATA

PDF Itinerary

JSON Data

CSV

Summer in Tokyo

Narita → Shibuya → Shinjuku

AUG 2024

>

Alpine Escapade

Zurich → Lucerne → Interlaken

DEC 2023

>

Parisian Weekend

Gare du Nord → Marais → Louvre

OCT 2023

>

View All Saved Trips

Security Tip

Enable two-factor authentication in the Admin panel for enhanced security of your trip data.

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